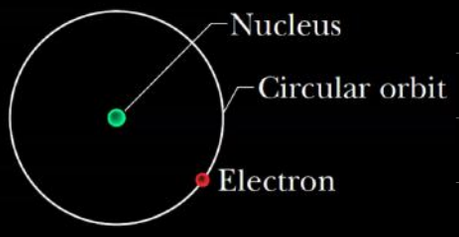
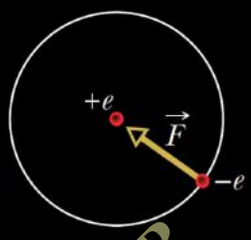
Bohr's Model Of Hydrogen Atom:





Recall from the previous chapter that how confinement of an electron in a potential well leads to quantization of it energies. The one dimensional trap for an electron was just an artificial one, however in nature, the atoms act like natural potential traps for electrons (due to attraction from nucleus). Various models were put forward for understanding the atomic structure (for example Rutherford model, sommerfeld's elliptical orbits model, Thomson Model etc about which you have studied in the atomic structure chapter in chemistry). Here we will be mainly dealing with Bohr's model. (We briefly discussed Rutherford's model in electrostatics). Bohr's model was proposed to explain discrete lines in the emission spectrum of hydrogen atom. Prior to Bohr, Balmer tried to give a formula for the emission lines in visible spectrum as:

$$\frac{1}{\lambda} = R\left[\frac{1}{2^2} - \frac{1}{n^2}\right] \quad \text{for} \quad n = 3, 4, 5 \text{ and } 6,$$

however it was a puzzle, why only four such wavelengths existed. Balmer was a mathematician who was just trying to fit a formula to the observed data. Soon after this Bohr realized that this formula could be derived if we could assume angular momentum to be quantized and thus gave his famous Quantization rule.

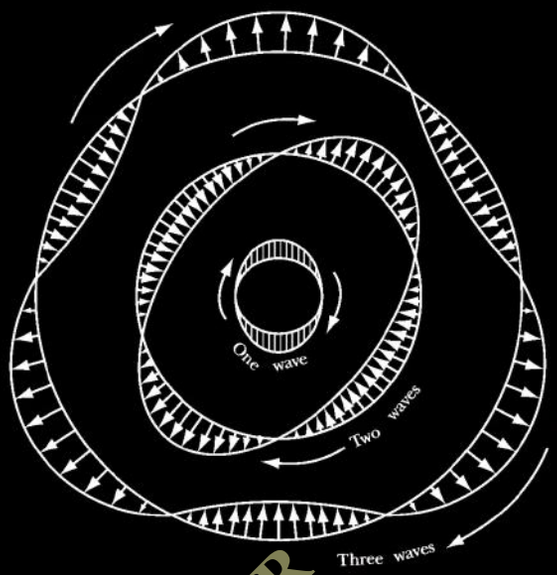
Bohr's Quantization Rule: According to Bohr electrons revolved around the nucleus in circular orbits due to electrostatic attraction and only those orbits were possible in which angular momentum was quantized, mathematically,

$$L = mvr = nh/2\pi \qquad n = 1, 2, 3 \ldots$$

m: mass of electron, v: speed of electron, r: radius of orbit.

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At that time Bohr did not know the reason for quantization, however later on quantization could be explained in terms of De Broglie waves (as was explained in the last chapter). Thus using the quantization rule, the assumption of circular orbits and conservation of electromechanical energy we will be able to derive various physical quantities of a Bohr Atom (Hydrogen like atom i.e single electron atoms / ions).



DERIVATION OF VARIOUS PHYSICAL QUANTITIES IN BOHR ATOM:

RADIUS OF nth BOHR ORBIT:

Suppose we have a single electron specie with nuclear charge Ze. Now balancing centrifugal and electrostatic forces:

$$\frac{1}{4\pi\epsilon_0}\frac{Ze^2}{r^2} = mv^2/r \quad (1)$$

By Quantization rule:

$$mvr = n\,h/2\pi \quad (2)$$

The only unknowns in these two equations are r and v and we can readily solve this for r, we get:

$$r_n = \frac{h^2\epsilon_0}{\pi m e^2}\frac{n^2}{Z} = 0.53\frac{n^2}{Z}\,\text{Å}$$

VELOCITY IN n^{th} BOHR ORBIT:

Solving equations (1) & (2) above, we get velocity in the nth orbit as:

$$v_n = \frac{e^2}{2\epsilon_0 h}\frac{Z}{n} = 2.165\times10^6\frac{Z}{n}\,ms^{-1}$$

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→ $2.16 \times 10^6\ ms^{-1} = c/137$, it is also called fine structure constant.

→ v_n is independent of mass of electron (unlike r_n) so even if electron were heavier or lighter v_n would be same).

Angular Velocity In n^{th} Bohr Orbit:

Once we know v_n and r_n, ω_n can be simply calculated as:

$$\omega_n = v_n / r_n = \frac{8\pi^3 K^2 e^4 m}{h^3} \frac{Z^2}{n^3} \qquad K: 1/4\pi\epsilon_0$$

Frequency In n^{th} Bohr Orbit:

We know that $\omega = 2\pi\nu$ thus,

$\nu = \omega/2\pi$ i.e

$$\nu_n = \frac{me^4}{4\epsilon_0^2 h^3} \frac{Z^2}{n^3} = 6.53 \times 10^{15} \frac{Z^2}{n^3}\ Hz$$

Time Period In n^{th} Bohr Orbit:

$$T_n = \frac{1}{f_n} = \frac{4\epsilon_0^2 h^3}{me^4} \frac{n^3}{Z^2} = 1.53 \times 10^{-16} \frac{n^3}{Z^2}$$

Current In n^{th} Bohr Orbit:

Recall that for charges orbiting in circle, effective current is charge x frequency thus:

$$I_n = e\nu_n$$

Magnetic Field At Nucleus Due To n^{th} Orbit:

$$B_n = \mu_0 I_n / 2r_n = \frac{8\pi^4 \mu_0 K^3 e^7 m^2}{h^5} \frac{Z^3}{n^5}$$

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MAGNETIC MOMENT OF nth BOHR ORBIT: Recall the gyromagnetic ratio for the spinning charges, $\frac{M}{L} = \frac{q}{2m}$ where M is the magnetic moment and L is the angular momentum i.e $nh/2\pi$. Thus we get

$$M_n = \frac{eh}{4\pi m} n$$

For $n=1$, this quantity is known as Bohr Magneton. Thus Bohr Magneton:

$$\beta = \frac{eh}{4\pi m} = 9.27 \times 10^{-24} \text{ J/T}$$

ENERGY IN n^{th} ORBIT: Total energy in n^{th} orbit is sum of Kinetic Energy and Potential Energy. Now:

$$KE_n = \frac{1}{2} m v_n^2 \quad \text{also} \quad m v_n^2 = \frac{K Ze^2}{r_n} \quad (\because \text{Coulomb force}).$$

Thus $KE_n = \frac{1}{2} \frac{K Ze^2}{r_n}$ also $PE = \frac{-K Ze^2}{r_n}$, thus total Energy,

$$E_n = KE_n + PE_n = -\frac{1}{2} \frac{K Ze^2}{r_n} \quad \text{thus} \quad E_n = -\frac{1}{2} \frac{ke^2 Z}{r_n}$$

From above analysis it can be readily seen that $E_n = -KE_n = \frac{PE_n}{2}$

Substituting the value of r_n, we get:

$$E_n = \frac{-me^4}{8\epsilon_0^2 h^2} \frac{Z^2}{n^2} = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

The value 13.6 eV is called 1 Rydberg Energy Unit.

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Energy Levels In Hydrogen Atom:



$E_1 = -13.6Z^2\,eV$

$E_2 = -3.40Z^2\,eV$

$E_3 = -1.51Z^2\,eV$

$E_4 = -0.85Z^2\,eV$

$E_5 = -0.54Z^2\,eV$

$E_6 = -0.36Z^2\,eV$

The energy values of some orbits are given in the table above, the student is advised to memorize the values because it helps in quickly recognizing the transition (otherwise it might involve a tedious integer equation involving $1/n^2 - 1/m^2$ where n and m are constrained to be integers).

Rydberg Constant: The ratio of Rydberg energy unit to the product hc is called Rydberg constant R. Thus:

$$R = 13.6\,eV/hc = |E_1|/hc = 10967800\,m^{-1}.$$

Very often in wavelength calculations we need reciprocal Rydberg constant:

$$\lambda_0 = 1/R \approx 911\,\text{Å} \quad (\text{Actually } 911.2671\,\text{Å}).$$

Wavelength Of A Transition:

If an electron jumps from m^{th} orbit to n^{th} orbit, a photon is emitted corresponding to difference of the energies in orbits. Thus:

$$h\nu = \frac{hc}{\lambda} = E_0\left[\frac{1}{n^2} - \frac{1}{m^2}\right] = Rhc\left[\frac{1}{n^2} - \frac{1}{m^2}\right]$$

$$\Rightarrow \frac{1}{\lambda} = R\left[\frac{1}{n^2} - \frac{1}{m^2}\right] \Rightarrow \boxed{\lambda = \frac{\lambda_0}{\left(\frac{1}{n^2} - \frac{1}{m^2}\right)}}$$

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Here λ_0 is reciprocal Rydberg constant (911 Å) which represents wavelength corresponding to a transition from $\infty \rightarrow 1$ (on photon of energy 13.6 eV).

DE BROGLIE WAVELENGTH IN n^{th} ORBITAL:

We have by Quantization rule $mvr = nh/2\pi$

$\Rightarrow \quad \lambda_D = \dfrac{h}{mv} = 2\pi r/n$, thus

$$\boxed{\lambda_D = \frac{2\pi r_n}{n}} \quad \text{Ans.}$$

EXAMPLE: If average life time of an excited state of hydrogen atom is 10^{-8} sec, estimate how many revolutions does an electron make when it is in state $n=2$ before transitioning to $n=1$?

Solution: No of orbits : $N = \omega \times t/2\pi = vt/2\pi r$

$$N = \frac{v_0(z/n) \times t}{2\pi r_0(n^2/z)} \quad n=2,\ z=1 \text{ thus, } N = \frac{v_0 t}{16\pi r_0}$$

$$= \frac{2.165\times10^6 \times 10^{-8}\ s^{-1}}{16\pi \times 0.53\times10^{-10}\ m} = 8.1\times10^6 \text{ revolutions Ans.}$$

EXAMPLE: What is angular momentum of electron in a hydrogen atom whose energy is 3.4 eV?

Solution: $E_n = 13.6/n^2 = 3.4 \Rightarrow n = 2$, $L = nh/2\pi = h/\pi$ Ans.

EXAMPLE: Electron in a hydrogen atom makes a transition $n_1 \rightarrow n_2$. The time period in initial state is 8 times the time period in final state. What are the possible values for n_1 and n_2?

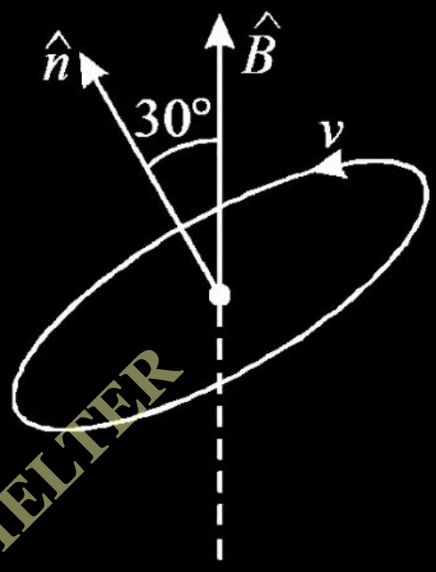
Sol: $T \propto 2\pi r/v$ i.e $T \propto (n^2/z)/(z/n)$ i.e n^3/z^2

Thus $n_1^3/n_2^3 = 8 \Rightarrow n_1/n_2 = 2$. All natural numbers

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satisfying this relation are part of the solution set.

EXAMPLE: An electron in ground state of hydrogen atom revolves anti-clockwise as shown. Find
(a) Orbital magnetic moment
(b) Torque if there is an angle of 30° between B and plane of rotation as shown.



Solution: $\frac{M}{L} = \frac{q}{2m} \Rightarrow M = \frac{q}{2m} L = \frac{q}{2m} \cdot \frac{nh}{2\pi}$

For ground state, $n = 1$, $\Rightarrow M = \frac{eh}{4\pi m}$ Ans.

Now $|\vec{\tau}| = |\vec{M} \times \vec{B}| = \frac{eh}{4\pi m} \times B \times \sin 150°$

$\Rightarrow \tau = ehB/8\pi m$ Ans.

EXAMPLE: Find the maximum wavelength that hydrogen atom can absorb in ground state, Also find next smaller wavelength.

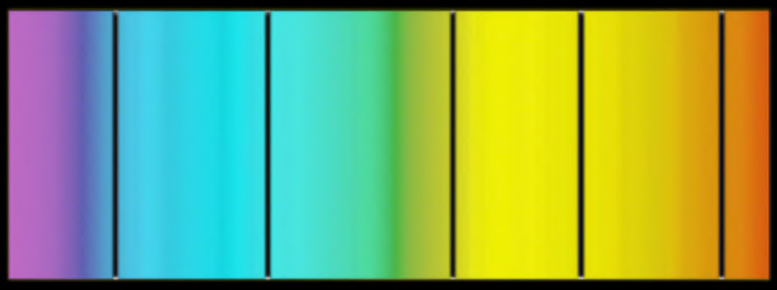
Solution: Maximum wavelength corresponds to least frequency i.e least energy. From ground state, least energy gap is 10.2 eV (1 → 2). Thus,

$\lambda = \frac{\lambda_0}{\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)} = \frac{911\,\text{Å}}{1 - 1/4} = 1214.7\,\text{Å}$ Ans.

For next smaller wavelength, we consider the transition 1 → 3, accordingly, wavelength: $\lambda = \frac{911\,\text{Å}}{1 - 1/9} \approx 1025\,\text{Å}$.

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Excitation And Ionization Energies:

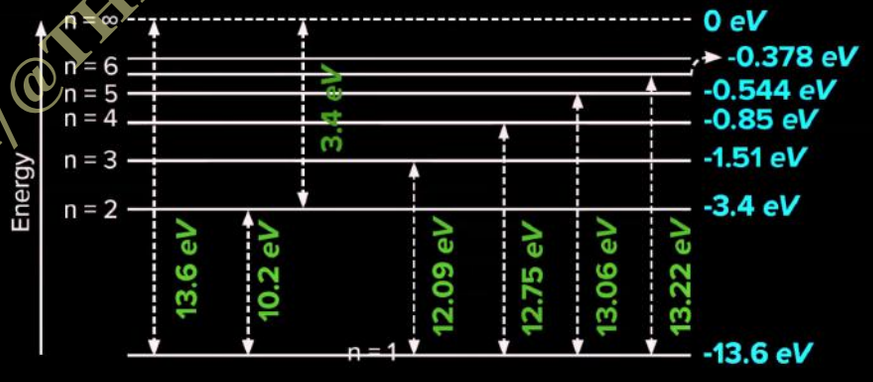


Absorption spectrum

To excite an electron from one quantum state to another, the photon must have exactly the energy equal to energy difference between two quantum states. If the photon has more energy than required for the transition, it is not absorbed, unless the energy is greater than the ionization energy. In case the photon energy is greater than the ionization energy ($1 \rightarrow \infty$), it can be absorbed by the electron and excess energy appears as kinetic energy of the electron. The transition energies for transition between any two states is given as:

$$\Delta E = 13.6\,eV\,Z^2\left[\frac{1}{n^2} - \frac{1}{m^2}\right]$$

The transition energies of some of the transitions for a hydrogen atom are listed out in the adjacent energy diagram. Students are advised to memorize the transition energies for a quick reckoning while solving the questions.



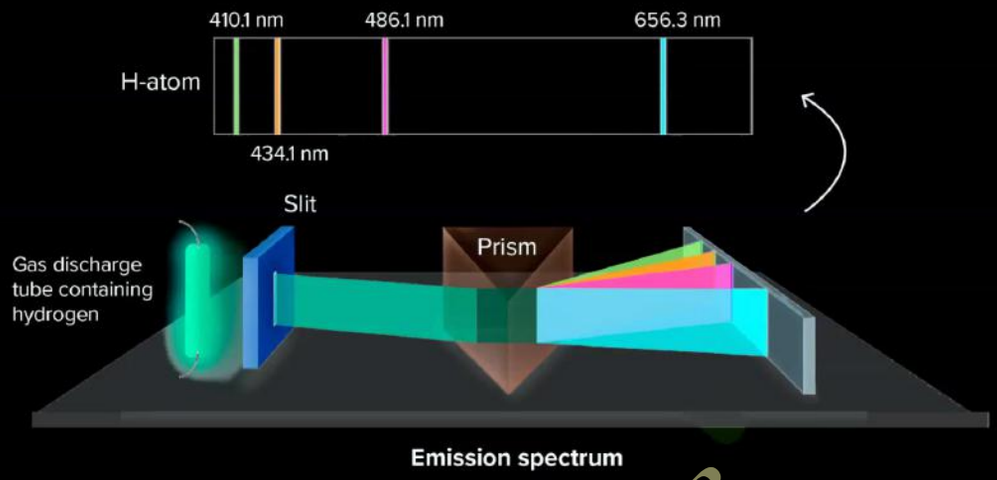
Energy level diagram for hydrogen atom

Absorption Spectrum: When light is made to pass through a tube containing hydrogen atom, in the emergent light certain wavelengths are missing. These correspond to transition from ground state i.e $1 \rightarrow n$. (Note: absorption spectrum always implies absorption from ground state unless explicitly mentioned otherwise).

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Emission Spectrum:

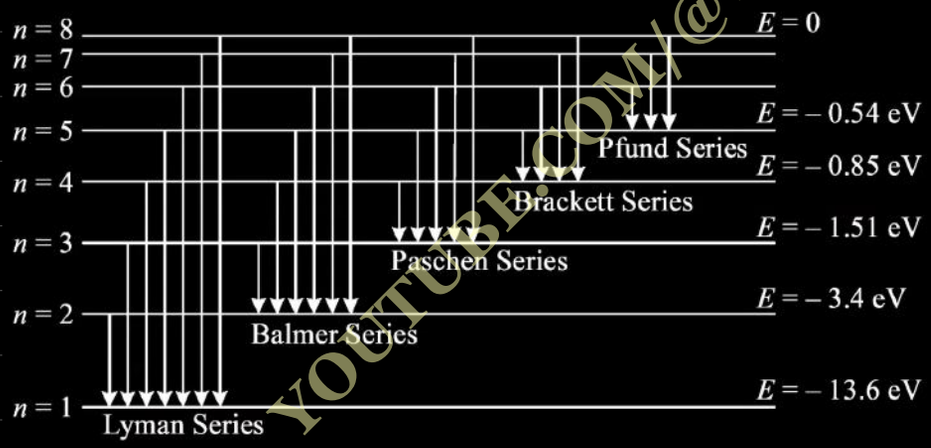
When, atoms of hydrogen gas are in an excited state in a discharge tube, random de-excitations occur between any two levels lying between n and 1. The wavelengths corresponding to these transitions correspond to the emission spectrum.

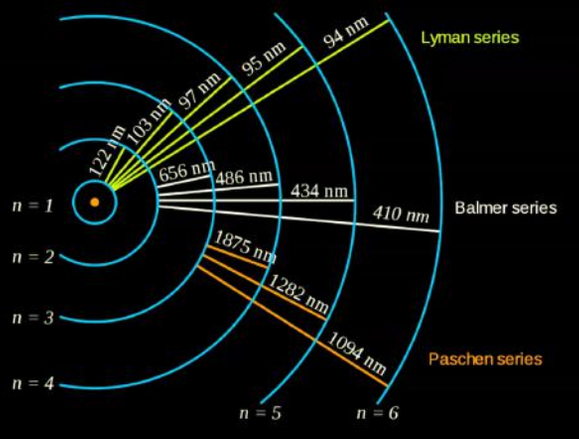


Since transitions can for choice of any two quantum numbers between 1 and n, thus number of lines in emission spectrum of a gas excited to greatest quantum number n is given as $N = C(n,2)$ i.e $n(n-1)/2$

$$N = n(n-1)/2$$

Hydrogen Spectrum Series:





Lyman Series: A transition of the form $n \rightarrow 1$ corresponds to wavelengths in Lyman series. These wavelengths lie in ultraviolet region of the Electromagnetic spectrum. First line is $2 \rightarrow 1$, second line is $3 \rightarrow 1$ and so on.

Balmer Series: A transition of the form $n \rightarrow 2$ corresponds to

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Balmer series. Four of its members lie in the visible corresponding to the transitions (3,2), (4,2), (5,2) and (6,2).

PASCHEN SERIES: A transition of the form $n \to 3$ forms the members of Paschen series, these lie in the near infrared region.

BRACKETT SERIES: A transition of the form $(n \to 4)$ forms the member of Brackett series. These members lie in the infrared region.

PFUND SERIES: A transition of the form $n \to 5$ corresponds to Pfund series. The members of the series lie in the deep infrared region.

EXAMPLE: The wavelength of the first member in Balmer series is 6563 Å, find the wavelength of first member in Lyman series.

Solution: $\lambda = \lambda_0 / (1/n^2 - 1/m^2)$, thus $6563 A^\circ = \lambda_0 / (1/4 - 1/9)$

also $\lambda_{21} = \lambda_0 / (1 - 1/4) \Rightarrow \lambda_{21} = 6563 A^\circ \left[\left(\frac{1}{4} - \frac{1}{9}\right) / (1 - 1/4) \right]$

$\Rightarrow \lambda_{21} = 1219.1 \text{ Å}$ Ans.

EXAMPLE: Find the ratio of ionization energy of Hydrogen atom and Li^{++}.

Solution: $E = E_0 Z^2/n^2$, thus for any atom ionization energy:

$$I.E. = E_0 Z^2 \left[\frac{1}{1} - \frac{1}{\infty} \right] = E_0 Z^2, \text{ thus}$$

$$E_H / E_{Li^{++}} = 1/3^2 = 1/9 \text{ Ans.}$$

EXAMPLE: Hydrogen atom in ground state is excited by radiation of wavelength 975 Å. How many different lines are possible in resulting emission spectrum. Also find longest wavelength in emission spectrum.

Solution: Energy corresponding to $975 A^\circ = \frac{12420}{975} = 12.74 eV \simeq 12.75 eV$

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This corresponds to the transition $1 \rightarrow 4$. Thus number of line: $4 \times (4-1)/2 = 6$ Ans.

Longest wavelength corresponds to least energy difference i.e $4 \rightarrow 3$.

Thus longest wavelength: $911\,\text{Å}\left(\frac{1}{9} - \frac{1}{16}\right)^{-1} = 18741\,\text{Å}$ Ans.

EXAMPLE: A hydrogen atom makes a transition from a state with binding energy 0.85 eV to state with excitation energy 10.2 eV. Characterize the transition (identify quantum numbers involved) and find the wavelength of emitted photon.

Solution: Binding energy is energy required to expel an electron to infinity. Thus binding energy in any state is simply $13.6\,Z^2/n^2$ eV.

Now $0.85 = 13.6/16$ thus, corresponding state is $n = 4$. Also, 10.2 eV is the excitation energy for $1 \rightarrow 2$ transition. Thus our transition is $4 \rightarrow 2$. Accordingly wavelength:

$$\lambda = 911\,\text{Å}\left[\frac{1}{4} - \frac{1}{16}\right]^{-1} = 4858\,\text{Å}$$ Ans.

EXAMPLE: A hydrogen like atom (atomic number Z) is in an excited state of quantum number n. It can make a transition to first excited state by successively emitting two photons of energies 10.2 eV and 17.0 eV respectively. Alternatively it can go to second excited state by emitting photons of energies 4.25 eV and 5.95 eV. Determine n and Z.

Solution: $n \rightarrow 2$ energy $= 10.2 + 17 = 27.2$ eV (first excited state is $n = 2$).

$n \rightarrow 3$ energy $= 4.25 + 5.95 = 10.2$ eV

$\Rightarrow$ $3 \rightarrow 2$ energy: $27.2 - 10.2 = 17$ eV $= 13.6\,Z^2\left[\frac{1}{4} - \frac{1}{9}\right]$

$$\Rightarrow Z^2 = \frac{17 \times 36}{13.6 \times 5} = 9 \Rightarrow Z = 3$$ Ans.

Now $13.6 \times 9\left[\frac{1}{4} - \frac{1}{n^2}\right] = 27.2 \Rightarrow \frac{1}{4} - \frac{1}{n^2} = \frac{2}{9} = \frac{1}{4} + \frac{2}{9} - \frac{1}{4}$

$$\Rightarrow \frac{1}{n^2} = \frac{1}{36} \Rightarrow n = 6$$ Ans.

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Excitation By Collision: Recall that maximum energy that can be lost in a two particle collision is $\frac{1}{2}\mu v_{rel}^2$ i.e KE available in CM frame (this is to ensure conservation of momentum). Thus to excite an electron through collision, the minimum Energy of the Projectile + Hydrogen atom system in CM frame must be equal to the least transition energy (10.2 eV from ground state). If the system has larger energy than this value, extra energy can appear as kinetic energy of the projectile and the hydrogen atom. Thus in case of a neutron on hydrogen atom colliding with another hydrogen atom at rest, the minimum energy from ground frame must be 20.4 eV. This can be shown as follows:

$$\frac{1}{2}\mu v^2 = 10.2\,eV \Rightarrow \frac{1}{2}\left(\frac{m}{2}\right) \times v^2 = 10.2\,eV \Rightarrow \frac{1}{2}mv^2 = 20.4\,eV$$

QED.

($\because \mu = mm/(m+m) = m/2$ for neutron on hydrogen atom as projectiles).

For electron as projectiles, reduced mass is almost same as electron mass because $m_e << m_H$. Thus min energy required by an electron to cause excitation is 10.2 eV itself.

Effect Of Finite Nuclear Mass:



So far in our analysis we assumed the nucleus to be stationary, however, actually both the electron and the nucleus will be orbiting about their common centre of mass. Let the distances from common centre of mass be A and a as shown in the figure above, then, if r is the separation between nucleus and electron, then:

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$$A = \left(\frac{m}{m+M}\right) r = \frac{\mu}{M} r \quad \text{and} \quad a = \left(\frac{M}{m+M}\right) r = \frac{\mu}{m} r$$

Also, if v is the velocity of electron relative to nucleus then, individual velocities are $\mu v/M$ and $\mu v/m$ ($\because v = p/m$).

Thus by angular momentum quantization:

$$\mu v r = \frac{nh}{2\pi} \quad ①$$

Also by centrifugal force balance, $\frac{kZe^2}{r^2} = m\left(\frac{\mu v}{m}\right)^2 \cdot \frac{1}{a}$

$$\Rightarrow \frac{kZe^2}{r^2} = \frac{\mu^2 v^2}{m} \cdot \frac{m}{\mu r} \quad (\because a = \mu r/m)$$

$$\Rightarrow \frac{kZe^2}{r^2} = \frac{\mu v^2}{r} \quad ②$$

Thus we see that equations ① & ② have same form as the equations with stationary nucleus except that now electronic mass is replaced by reduced mass. This will get carried in to solution as well, thus all results we obtained with stationary nucleus are still valid except that we need to replace m by μ.

→ Note that $\frac{1}{2}\mu v^2$ here represents combined KE of electron + nucleus this will be split in inverse ratio of masses between electron and nucleus.

→ For purpose of wavelength calculations, we need to revise Rydberg's constant considering the mass of nucleus, Recall that Energy is proportional to m^1 and is equal to Rhc thus we will need to modify R as:

$$R = R_\infty (\mu/m) = \frac{R_\infty M}{M+m}$$

$$\Rightarrow R = \frac{R_\infty}{1 + m/M}$$ where R_∞ is Rydberg constant assuming nucleus of infinite mass.

Also, accordingly $\lambda_0' = \lambda_0 [1 + m/M]$ thus actual wavelengths in transitions will be longer.

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→ Isotopes have same nuclear charge but different nuclear masses so they will have a slightly different spectrum. Heavier isotopes will emit smaller wavelengths.

→ For a ready reckoning, mass proportionality of various physical quantities are mentioned in adjacent table. It is worth mentioning that v is independent of m while radius is proportional to m^{-1}.

	r	v (speed)	E	$v(f)$	R
Case I	$1/m_e$	independent	m_e	m_e	m_e
Case II	$1/\mu$	independent	μ	μ	μ
	$\frac{n^2}{Z}$	$\frac{Z}{n}$	$\frac{Z^2}{n^2}$	$\frac{Z^2}{n^3}$	Z^2
	h^2	$1/h$	$1/h^2$	$1/h^3$	$1/h^3$

r: Nucleus to Electron distance

If we consider nucleus to be moving we need to substitute m by μ.

→ v is the speed of electron relative to nucleus, for individual speed we need to split it in inverse ratio of masses.

EXAMPLE: A muon is an unstable elementary particle with charge either $+e$ or $-e$. A negative muon can be captured by a proton to form a muonic atom (a) Find the muon to proton distance for this atom (b) Find ionization energy of this atom, given muon is 207 times heavier than electron.

Solution: $r \propto m^{-1}$ thus $r_{muon} = r_0\left(\frac{m}{\mu}\right)$, $\mu = \frac{207m_e \times 1836m_e}{207m_e + 1836m_e}$

$\Rightarrow \mu = 186\,m_e$, thus $r_{mu} = 0.53 \times \frac{1}{186}\,\text{Å} = 2.85 \times 10^{-13}\,m$ Ans.
$= 285\,fm$.

Now energy $\propto m^1$ thus ionization energy for muonic atom:
$IE = 13.6 \times 186\,eV = 2.53\,keV$ Ans.

→ Blunder Alert: Remember to calculate reduced mass of muon. Don't simply replace electron mass by muon mass because its mass is comparable to proton mass. (Had it been much lighter say twice electron mass then reduced mass would also have been almost double).

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→ Velocity relative to nucleus will be same as that in ordinary hydrogen atom but relative to centre of rotation it must be split in inverse ratio of masses.

EXAMPLE: Electrons and positrons (e^+) can form a bound system called positronium. (a) What is the first excitation energy of the positronium (b) What is the radius of ground state of positronium.

Solution: $\mu = \frac{m \times m}{m+m} = m/2$. Now excitation energy:

$$\Delta E = 13.6\,eV \times \frac{\mu}{m}\left[1 - \frac{1}{4}\right] = 5.1\,eV \quad \text{Ans.}$$

$$r = 0.53\,\text{Å} \times \frac{m}{\mu} = 1.06\,\text{Å} \quad \text{Ans.} \quad (\text{This is electron to positron distance}).$$

The individual radii of electron and positron are half of this value (about common CM).

EXAMPLE: Consider a hypothetical hydrogen atom in which potential energy between a proton electron pair at separation r is given as: $U = k \ln r - k/r$ where k is a constant. For such a hypothetical atom, calculate the Bohr radii and energy levels.

Solution: $F = -\partial U/\partial r = k/r = mv^2/r$ (− sign signifies attraction)

$$\Rightarrow \quad mv^2 = k \quad (1)$$

and $mvr = nh/2\pi$ (2)

Solving (1) & (2), $r = \frac{nh}{2\pi\sqrt{mk}}$ Ans $\qquad v = \sqrt{\frac{k}{m}}$

Now energy $= \frac{1}{2}mv^2 + U = \frac{k}{2} + k \ln \frac{nh}{2\pi\sqrt{mk}} - k/2$

$$\Rightarrow \quad E_n = k \ln \frac{nh}{2\pi\sqrt{mk}} \quad \text{Ans.}$$

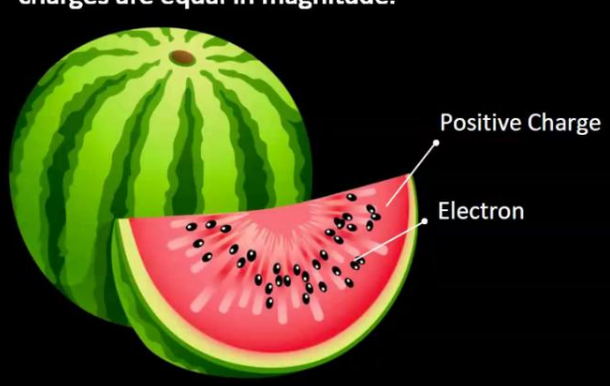
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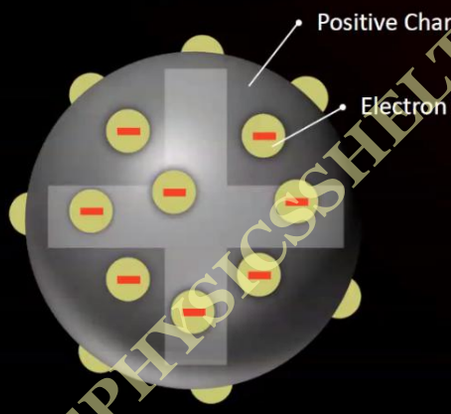
Thomson's Plum And Pudding Model :

Thomson's plum and pudding model was introduced to explain thermionic emission of electrons. However it was unable to explain the deviation of alpha particles in Rutherford's scattering experiment. (According to Thomson's Model, average deflection would be proportional to square root of thickness of Gold foil while actually it is proportional to thickness of the foil).

Key Points In Thomson's Model :

- An atom consists of a positively charged sphere with electrons embedded in it.
- An atom as a whole is electrically neutral because the negative and positive charges are equal in magnitude.



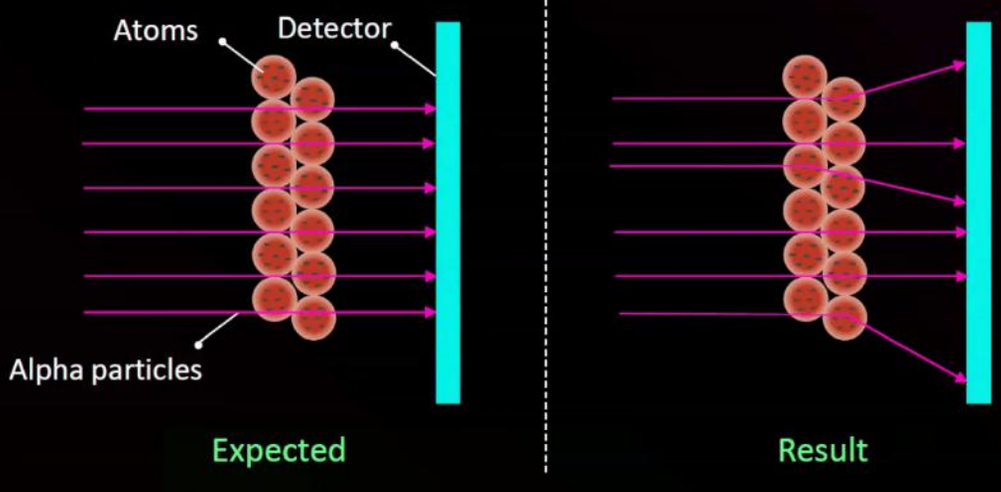


- According to J. J. Thomson, an atom was a solid sphere of positively charged material and the negatively charged particles, the electrons were embedded in it, like the seeds of the watermelon. The atomic model is compared to watermelon, where he considered the red part of the watermelon as positively charged and watermelon seeds as negatively charged particles, embedded in it.

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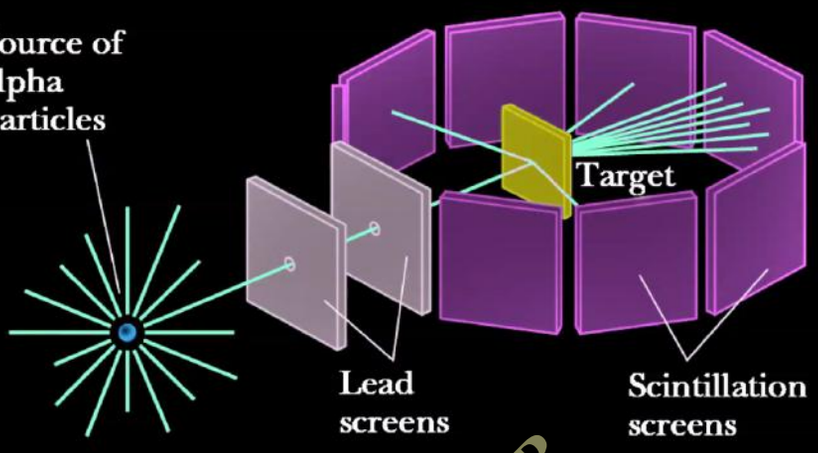
Limitations :

- Failed to explain the stability of an atom.
- Failed to explain how a positive charge holds negatively charged electrons in an atom.
- Failed to explain the scattering of alpha particles by thin foils.



Rutherford's Alpha Particle Scattering:

Rutherford and his students Geiger and Marsden conducted experiments by bombarding a thin gold foil with alpha particles from a radioactive source. A schematic of their apparatus is shown in adjacent figure.

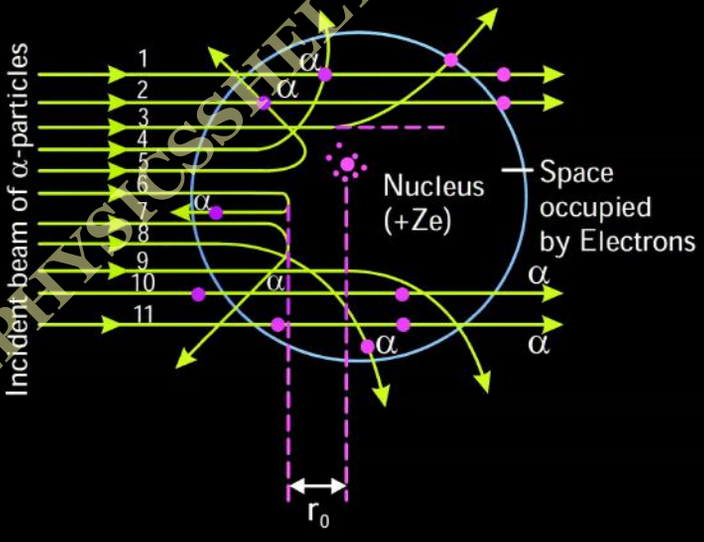


Observations:

→ Most alpha particles passed through the gold foil undeviated thus implying that most of the atom was empty space

→ Some of particles deviated through angles between 0 and 90°

→ Very few of particles were found to deviate by angles more than 90° (In fact at first attempt Rutherford could not detect any back scattering. Later on his students performed the experiments with improved setups to detect back-scattering).



→ An extremely small number of particles (about 1 in 8000) practically retraced their paths i.e suffered a deviation of nearly 180°.

Conclusions:

From above observations, following conclusions can be made:

→ Most portion of the atoms is hollow space

→ The positive charge of the atoms must be concentrated in a very small space at the centre of the atom (known as nucleus) which is responsible for causing very large deviations of extremely small number of particles.

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Distance Of Closest Approach:

If an alpha particle is approaching the nucleus Head-On, its distance of closest approach can be found by conservation of electromechanical energy. Thus if mass of α particle is m_α and velocity v_α, by energy conservation:

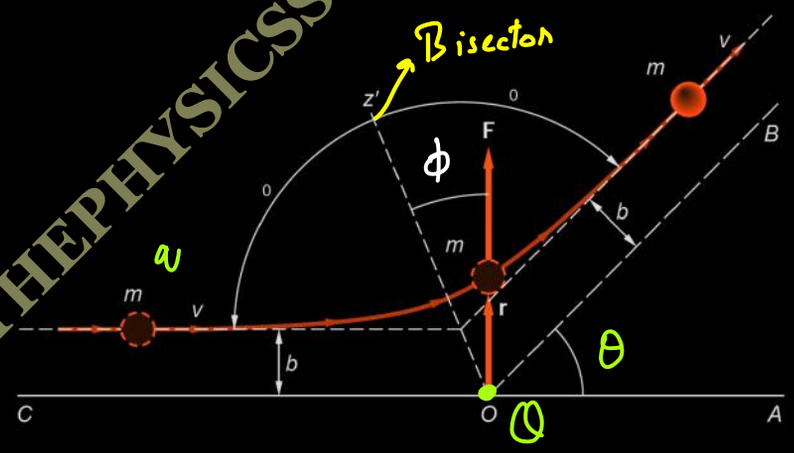
$$\frac{1}{2} m_\alpha v_\alpha^2 = \frac{1}{4\pi\varepsilon_0}\left(\frac{Ze \times 2e}{r_0}\right) \qquad (r_0: \text{Closest distance})$$

Thus,

$$\boxed{r_0 = \frac{1}{4\pi\varepsilon_0}\left[\frac{4Ze^2}{m_\alpha v_\alpha^2}\right]}$$

Angle Of Deviation:

We derived the formula for angle of deviation in Rutherford Model during our study of electric potential (student is advised to have a look at the derivation). For an impact parameter b, the Deviation angle Θ is given by the equation:



$$\boxed{\tan(\Theta/2) = \frac{kQq}{2Tb}}$$

when: $k: \frac{1}{4\pi\varepsilon_0}$, Q: nuclear charge

q: charge of projectile T: KE of projectile and b: impact parameter.

Number Of Particles Per Unit Detector Area:

The number of particles per unit detector area for a given scattering angle can be calculated using the relation between impact parameters and deviation angle and some geometry. The formula for number of particles per unit detector area is given as:

$$\boxed{\frac{dN}{dA} = \frac{N_i n L Z^2 k^2 e^4}{4r^2 T^2 \sin^4(\Theta/2)}}$$

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Here: N_i: Total number of incident particles
n: Atoms/unit volume in target
L: Thickness of target
Z: Atomic number of Target
e: Electron charge

k: $1/4\pi\epsilon_0$
r: Target to detector distance
T: Kinetic Energy of α particles
θ: Scattering angle

Here we can intuitively understand the significance of some of the variables. We see that:

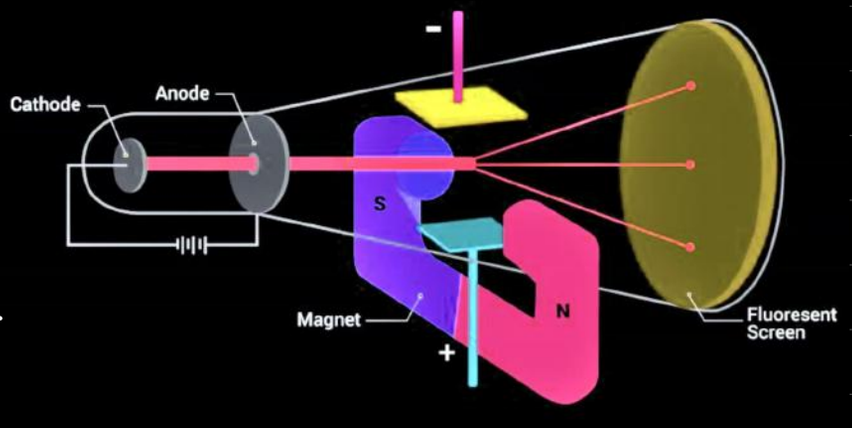
→ dN/dA is proportional to thickness L because almost each layer of atoms has same number of incident α particles (because nearly all α particles pass through), so more the number of layers crossed, more the number of deviated particles.

→ dN/dA is proportional to $1/r^2$ due to spherical symmetry as r increases same number of particles spread out on a larger sphere.

→ Similarly dN/dA is proportional to particle density. Closer the particles of gold foil, more the likelihood of deviation.

J.J. Thomson's Experimental Determination Of e/m Ratio:

In this experiment a beam of electrons is accelerated through some potential difference and made to pass through plates of a capacitor and poles of a magnet. The Electric and Magnetic fields are applied $\perp$ to the beam direction and their values are so adjusted that net Lorentz force on electrons in the apparatus is zero. Thus:



$$E = vB \quad (1) \qquad (\because E \perp v \perp B)$$

Also if V_0 is accelerating potential difference:

$$\frac{1}{2}mv^2 = eV_0 \quad (2)$$

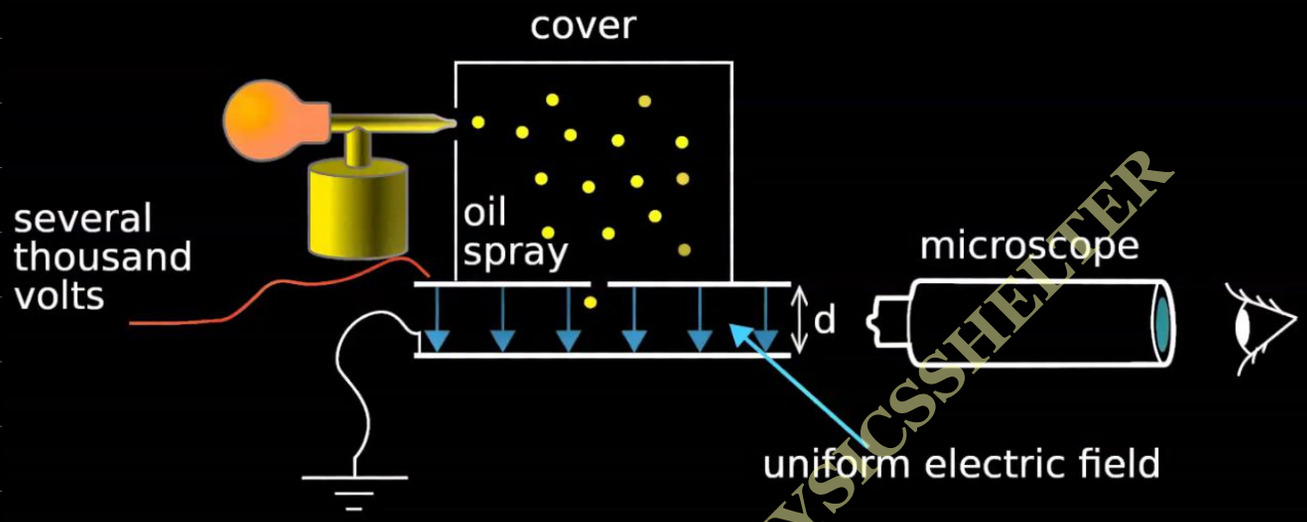
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eliminating v between ① & ②:

$$\frac{1}{2} m \left(\frac{E}{B}\right)^2 = eV_0$$

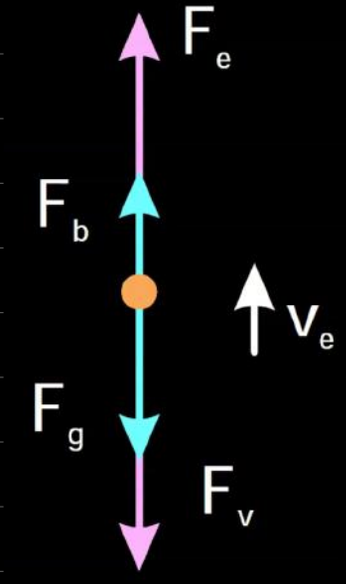
$$\Rightarrow \boxed{e/m = E^2/2V_0B^2}$$

MILLIKAN'S OIL DROP EXPERIMENT: CHARGE QUANTIZATION:



In Millikan's Oil drop experiment, oil drops are sprayed between two capacitor plates. The oil drops are charged by irradiating with ultraviolet light. Their charges are estimated by finding their terminal velocities under action of gravity and combined action of gravity and electric field. After finding charges on several oil drops, the HCF of the charges is found to calculate the value of minimum possible quantized charge. The steps and calculations in the experiment are as follows:

→ First electric field is adjusted until the drop begins to move upwards slowly (if the drop is negatively charged, upward plate must be positive for this to happen). The drop is allowed to achieve terminal velocity and the terminal velocity is calculated by observing the drop through the microscope. By force balance, if v_e is the terminal velocity in this case:



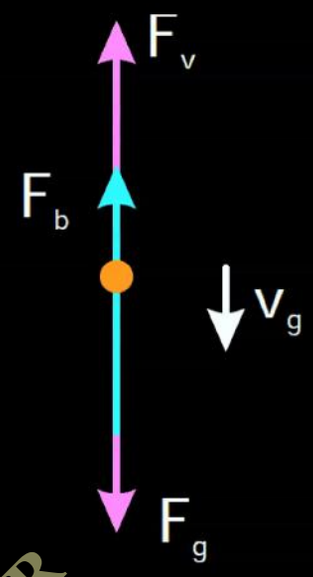
$$qE + \frac{4}{3}\pi R^3 \rho_a g = 6\pi\eta R v_e + \frac{4}{3}\pi R^3 \rho g \quad ①$$

ρ_a: Density of air, ρ: density of oil, η: viscosity of air.

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→ Now electric field is switched off and drops are allowed to attain a new downward terminal velocity under action of gravity. Thus if v_g be the downward terminal velocity then by force balance:

$$\frac{4}{3}\pi R^3 \rho_a g + 6\pi\eta R v = \frac{4}{3}\pi R^3 \rho g \quad (2)$$



Now (1) − (2) gives:

$$qE - 6\pi\eta R v_g = 6\pi\eta R v_e \quad \text{thus:}$$

$$q = 6\pi\eta R (v_e + v_g)/E$$

Now if we want we can substitute R from equation (2)

Thus we get:

$$\boxed{q = \frac{6\pi\eta (v_e + v_g)}{E}\left[\frac{9\eta v_g}{2(\rho - \rho_a)}\right]^{1/2}}$$

→ In our derivation, we took v_e as upwards and v_g as downwards, we can do similar derivations by taking v_e downwards also

Example: A charged oil drop of charge q is falling with terminal velocity v under action of gravity, in absence of electric field. Now an electric field is applied to make the drop stationary. If the drop acquires some additional charge, it moves up with a velocity 3v. Find new charge on the drop. (Ignore Buoyant Force).

Solution: Let the air drag force be kv where k is a proportionality constant and v the speed of the drop. Then, by force balance, initially:

$$mg = kv \quad (1) \qquad \Rightarrow mg = kv = qE$$

also after field is on, $mg = qE$ (2)

Now let new charge be q' then by force balance:

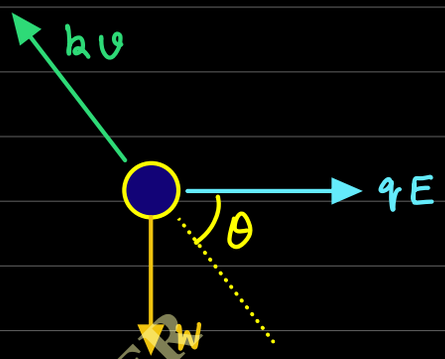
$$mg + 3kv = q'E \quad (3)$$

$$\Rightarrow \quad qE + 3qE = q'E \quad \Rightarrow \quad q' = 4q \text{ Ans.}$$

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Example: In Millikan's oil drop experiment, an oil drop of weight W is subjected to a horizontal electric field E. It moves with a constant velocity at an angle θ with horizontal. Find the charge on the drop.

Solution: Balancing Forces by Lami's theorem,

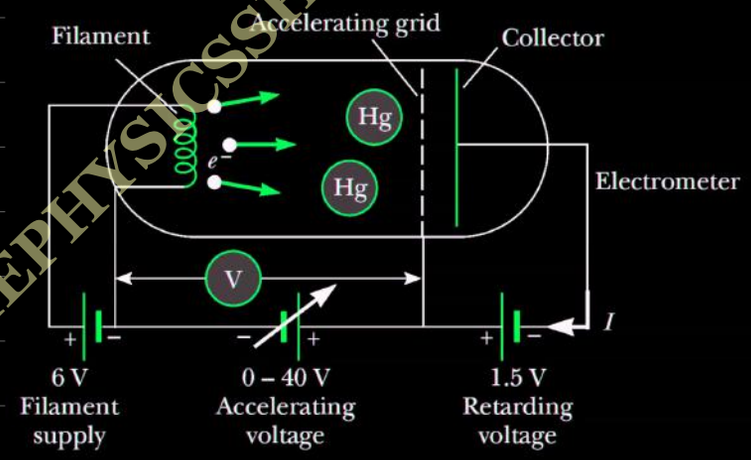


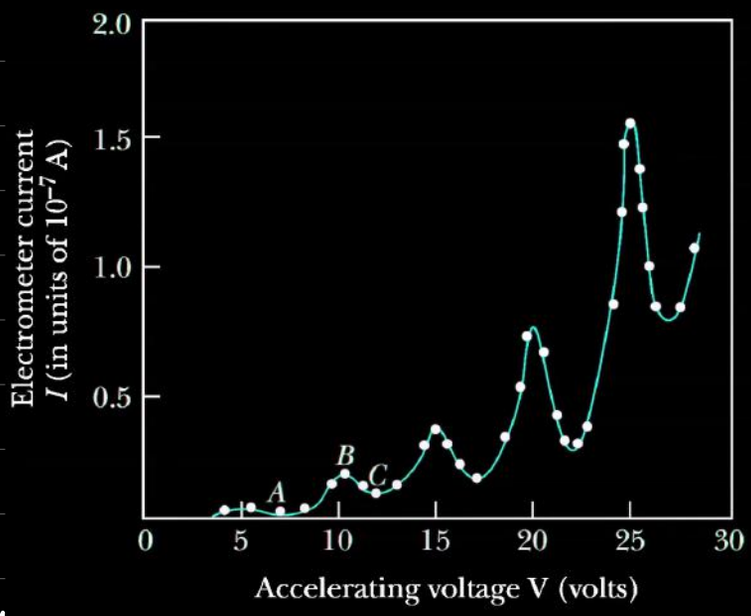
$$W / \sin(180° - \theta) = qE / \sin(90° + \theta)$$

$$\Rightarrow q = W \cot\theta \quad \text{Ans.}$$

Franck Hertz Experiment: Direct Confirmation Of Atomic Energy Levels:

In Franck Hertz experiment, electrons are accelerated by the applied voltage up to accelerating grid and after that they are decelerated while traveling from grid to collector due to reverse potential difference. Only those electrons are able to reach the collector, which have an an energy more than the negative work done by Retarding Voltage. We keep on increasing the accelerating voltage and we see a monotonic increase in current until a point where the electron energies are sufficient to excite mercury atoms to next energy level. When some electrons excite the mercury atoms, they lose kinetic energy and they will not have enough energy to cross the region of retarding potential, thus we see a sudden dip in current. Similar dips are also observed whenever accelerating potential is close to any excitation potential of mercury atoms. Thus confirming





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the existence of discrete energy levels.

EXAMPLE: PATHFINDER: MCQ 4:

What is principal quantum number of the highest energy level for which the lines in the Balmer series of emission spectra of atomic hydrogen will just be resolved by a spectrometer of resolving power $\lambda/d\lambda = 8 \times 10^3$?

(a) 30 (b) 40
(c) 50 (d) 60

Solution: Balmer transitions are of the form $n \to 2$ thus to distinguish between two nearby levels if λ is wavelength for $n \to 2$ transition and $\lambda - \delta\lambda$ for $(n+1)$ to 2 transition then for differentiating between these two nearby lines, $\lambda/\delta\lambda = 8 \times 10^3$

Now:

$$\frac{1}{\lambda} = R\left[\frac{1}{4} - \frac{1}{n^2}\right] \quad (1)$$

$$\frac{1}{\lambda - \delta\lambda} = R\left[\frac{1}{4} - \frac{1}{(n+1)^2}\right] \quad (2)$$

$(1)/(2) \Rightarrow$ $$1 - \frac{\delta\lambda}{\lambda} = \left(\frac{1}{4} - \frac{1}{n^2}\right) \Big/ \left[\frac{1}{4} - \frac{1}{(n+1)^2}\right]$$

$\Rightarrow$ $$\frac{\delta\lambda}{\lambda} = \frac{\left[\frac{1}{n^2} - \frac{1}{(n+1)^2}\right]}{\frac{1}{4} - \frac{1}{(n+1)^2}} = 1/8000$$

$\Rightarrow$ $$\frac{4\left[(n+1)^2 - n^2\right]}{(n+1)^2(n^2) - 4n^2} = \frac{1}{8000} \quad (1)$$

Solving we get $n \approx 39.5$ $\therefore$ Best option is (b).

Approximate solving method: Assuming $n >> 1$, we will just find the highest order terms in numerator and denominator of (1) thus:

$$\frac{4(2n)}{n^4} \approx \frac{1}{8000} \Rightarrow n^3 \approx 64000$$
$$\Rightarrow n \approx 40 \text{ ans.}$$

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EXAMPLE: PATHFINDER: MCQS:

A moving neutron collides with a singly ionized stationary helium atom that is in ground state. What should be the minimum speed of the moving neutron for the collision to be perfectly inelastic?

(a) 2.50×10^4 m/s (b) 4.25×10^4 m/s
(c) 6.25×10^4 m/s (d) 9.89×10^4 m/s

Solution: Minimum energy in CM frame must be equal to excitation energy of Helium from $1 \rightarrow 2$, Thus $\frac{1}{2}\mu v^2 = E_0 Z^2\left[1-\frac{1}{4}\right]$

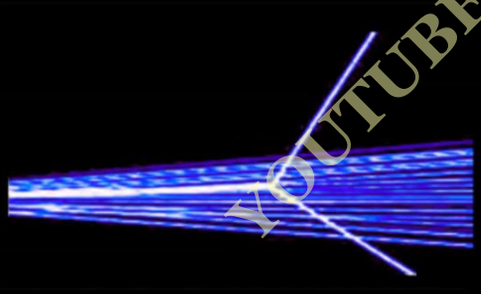
$\Rightarrow \frac{1}{2}\times\frac{4m}{5}v^2 = 13.6\,eV \times 4 \times 3/4$, Now $m = 931.5\,MeV/c^2$

Thus $\Rightarrow \frac{2}{5}\times 931.5\,MeV \times \frac{v^2}{c^2} = 13.6 \times 3\,eV$

$\Rightarrow \frac{v}{c} = \sqrt{\frac{13.6\times 3\times 5}{2\times 931.5\times 10^6}} = 3.31\times 10^{-4}$

$\Rightarrow v = 3\times 10^8 \times 3.31\times 10^{-4}\ m/s = 9.9\times 10^4\ ms^{-1}$, thus best answer is (d)

EXAMPLE: PATHFINDER: MCQ7:



A stream of α-particles encounters a stationary unknown nucleus in a cloud chamber. After a collision, the two nuclei are scattered in mutually perpendicular directions. If some kinetic energy is lost in the collision, the unknown nucleus must be

(a) a proton (b) a nucleus lighter than an α-particle
(c) an α-particle (d) a nucleus heavier than an α-particle

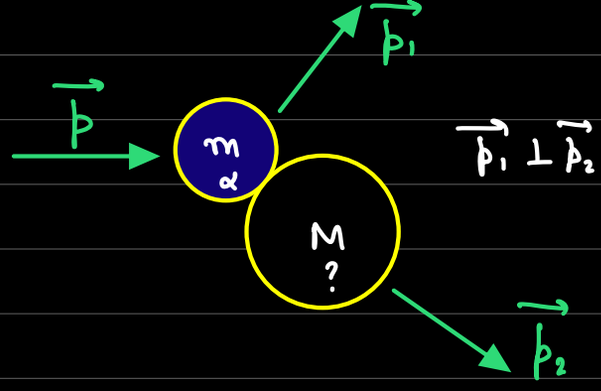
Solution: By momentum conservation:

$$p^2 = p_1^2 + p_2^2 \quad (1)$$

Also since some energy is lost (in excitation)

Thus:

$$\frac{p^2}{2m} > \frac{p_1^2}{2m} + \frac{p_2^2}{2M} \quad (2)$$



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(2) − (1)/2m ⇒ :

$$0 > \frac{p_2^2}{2}\left[\frac{1}{M} - \frac{1}{m}\right] \Rightarrow \frac{1}{m} > \frac{1}{M} \Rightarrow M > m$$

Thus alpha particle must have collided with a particle of mass lesser than its own. ∴ option (d) Ans.

EXAMPLE: PATHFINDER: BUILD: 8

If a gaseous sample consisting of identically excited hydrogen like ions is irradiated by a stream of photons each of energy $144E_0/225$, ten distinct lines are observed in the emission spectrum. Some of the lines correspond to energy lower than, some higher than and some equal to the energy of the incident photons. Here E_0 is the ionization energy of a hydrogen atom in ground state.

(a) In which excited state were the ions in the gaseous sample before interaction with the incident photons?
(b) To which highest state were the ions in the gaseous sample excited?
(c) Identify the element in the gaseous sample.

Solution: Ten distinct lines in emission spectrum implies that ions must have excited to $n = 5$ (∵ $C(5,2) = 10$). Let initial quantum level of the ions be m then according to the problem:

$$E_0 Z^2\left[\frac{1}{m^2} - \frac{1}{25}\right] = \frac{144\,E_0}{225}$$

$$\Rightarrow Z = \frac{12}{15}\Big/\sqrt{\frac{1}{m^2} - \frac{1}{25}}$$ Now ∵ Z is an integer, the term in square root must be a perfect square for one of the values of m from 1 to 4. By trial and error, we get integer value of $m = 3$ which makes the term is square root $\frac{16}{25\times 9}$

Thus $Z = \frac{12}{15} \times \frac{5\times 3}{4} = 3$ Thus ion is Li^{++}, $m = 3$, $n = 5$ Ans.

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